Parametric Equations

ANSWER KEY

Section 1: Plotting and Direction of Motion

1 For the parametric curve $x = 2t + 1, y = t^2 - 4$ for $-2 \le t \le 2$, find the initial point and terminal point.

a. **Initial:** $(-3, 0)$**, Terminal:** $(5, 0)$ ✓

b. Initial: $(1, -4)$, Terminal: $(5, 0)$

c. Initial: $(-3, 0)$, Terminal: $(1, -4)$

d. Initial: $(0, -3)$, Terminal: $(0, 5)$

At $t = -2$: $x = 2(-2) + 1 = -3, y = (-2)^2 - 4 = 0$. At $t = 2$: $x = 2(2) + 1 = 5, y = 2^2 - 4 = 0$.

2 How is orientation (direction of motion) shown on a parametric graph?

a. **Arrows placed along the curve indicating the direction as the parameter t increases** ✓

b. By the slope of the curve

c. By the color of the line

d. By the quadrant of the initial point

Orientation arrows indicate the forward direction of motion as time/parameter t advances.

3 Determine the direction of motion for $x = 3 - t, y = 2t$ as t increases.

a. **Leftward and upward** ✓

b. Rightward and upward

c. Leftward and downward

d. Rightward and downward

As t increases, $x = 3 - t$ decreases (leftward) and $y = 2t$ increases (upward).

Section 2: Eliminating the Parameter (Algebraic)

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4 Eliminate the parameter t from $x = t - 3, y = 2t + 1$.

a. $y = 2x + 7$ ✓

b. $y = 2x - 5$

c. $y = 2x - 7$

d. $y = x + 4$

Solve for t: $t = x + 3$. Substitute into y: $y = 2(x + 3) + 1 = 2x + 6 + 1 = 2x + 7$.

5 Eliminate the parameter from $x = t^2, y = 3t$ for $t \geq 0$.

a. $x = \frac{y^2}{9}$ for $y \geq 0$

b. $y = 3\sqrt{x}$

c. **Both of the above are correct** ✓

d. $x = 9y^2$

$t = y/3 \Rightarrow x = (y/3)^2 = y^2/9$, or equivalently $y = 3\sqrt{x}$ for $x \geq 0, y \geq 0$.

6 Eliminate the parameter from $x = e^t, y = 2e^{2t} - 1$.

a. $y = 2x^2 - 1$ **for** $x > 0$ ✓

b. $y = 2x - 1$

c. $y = e^{2x} - 1$

d. $y = 4x^2 - 1$

$e^{2t} = (e^t)^2 = x^2$. Since $x = e^t > 0$, $y = 2x^2 - 1$ with domain $x > 0$.

Section 3: Eliminating the Parameter (Trigonometric)

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7 Eliminate the parameter from $x = 3\cos\theta, y = 3\sin\theta$.

a. $x^2 + y^2 = 9$ ✓

b. $x^2 + y^2 = 3$

c. $9x^2 + 9y^2 = 1$

d. $\frac{x^2}{9} - \frac{y^2}{9} = 1$

$\cos\theta = x/3, \sin\theta = y/3$. Using $\cos^2\theta + \sin^2\theta = 1$: $(x/3)^2 + (y/3)^2 = 1 \Rightarrow x^2 + y^2 = 9$.

8 Identify the Cartesian equation for $x = 4\cos\theta, y = 2\sin\theta$.

a. $\frac{x^2}{16} + \frac{y^2}{4} = 1$ **(an ellipse)** ✓

b. $x^2 + y^2 = 20$

c. $\frac{x^2}{4} + \frac{y^2}{16} = 1$

d. $\frac{x^2}{16} - \frac{y^2}{4} = 1$

$(x/4)^2 + (y/2)^2 = \cos^2\theta + \sin^2\theta = 1 \Rightarrow \frac{x^2}{16} + \frac{y^2}{4} = 1$.

9 What curve is traced by $x = 3\sec\theta, y = 2\tan\theta$?

a. **A hyperbola:** $\frac{x^2}{9} - \frac{y^2}{4} = 1$ ✓

b. An ellipse: $\frac{x^2}{9} + \frac{y^2}{4} = 1$

c. A parabola: $y = 2x^2$

d. A circle: $x^2 + y^2 = 13$

Using the identity $\sec^2\theta - \tan^2\theta = 1$: $(x/3)^2 - (y/2)^2 = 1 \Rightarrow \frac{x^2}{9} - \frac{y^2}{4} = 1$.

Section 4: Projectile Motion Applications

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10 State the standard parametric equations for projectile motion launched with speed v_0 at angle θ from $(0, h_0)$ (with $g = 32$ ft/s^2).

a. $x(t) = (v_0\cos\theta)t, y(t) = -16t^2 + (v_0\sin\theta)t + h_0$ ✓

b. $x(t) = v_0 t, y(t) = -32t^2 + h_0$

c. $x(t) = (v_0\sin\theta)t, y(t) = -16t^2 + h_0$

d. $x(t) = (v_0\cos\theta)t, y(t) = (v_0\sin\theta)t$

Horizontal velocity is constant ($v_0\cos\theta$), and vertical position is governed by gravity: $y(t) = -\frac{1}{2}gt^2 + v_{0y}t + h_0$.

11 A ball is kicked from ground level at 64 ft/s at an angle of 30°. At what time does it hit the ground?

a. 2 seconds ✓

b. 1 second

c. 4 seconds

d. 3 seconds

$v_{0y} = 64\sin(30^\circ) = 32$ ft/s. $y(t) = -16t^2 + 32t = -16t(t-2) = 0 \Rightarrow t = 2$ s.

12 In the previous question, what is the horizontal range of the ball?

a. $64\sqrt{3}$ ft ✓

b. 128 ft

c. $32\sqrt{3}$ ft

d. 64 ft

$x(t) = (64\cos 30^\circ)t = (32\sqrt{3})(2) = 64\sqrt{3}$ ft.

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